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RESEARCH PAPER

Poly A tail length analysis of in vitro transcribed mRNA by LC-MS

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Abstract

The 3′-polyadenosine (poly A) tail of in vitro transcribed (IVT) mRNA was studied using liquid chromatography coupled to mass spectrometry (LC-MS). Poly A tails were cleaved from the mRNA using ribonuclease T1 followed by isolation with dT magnetic beads. Extracted tails were then analyzed by LC-MS which provided tail length information at single-nucleotide resolution. A 2100-nt mRNA with plasmid-encoded poly A tail lengths of either 27, 64, 100, or 117 nucleotides was used for these studies as enzymatically added poly A tails showed significant length heterogeneity. The number of As observed in the tails closely matched Sanger sequencing results of the DNA template, and even minor plasmid populations with sequence variations were detected. When the plasmid sequence contained a discreet number of poly As in the tail, analysis revealed a distribution that included tails longer than the encoded tail lengths. These observations were consistent with transcriptional slippage of T7 RNAP taking place within a poly A sequence. The type of RNAP did not alter the observed tail distribution, and comparison of T3, T7, and SP6 showed all three RNAPs produced equivalent tail length distributions. The addition of a sequence at the 3′ end of the poly A tail did, however, produce narrower tail length distributions which supports a previously described model of slippage where the 3′ end can be locked in place by having a G or C after the poly nucleotide region.

Keywords mRNA · Poly A · Tail length · Mass spectrometry · Slippage · In vitro transcription

Introduction

Eukaryotic mRNAs often terminate with a stretch of adenosines on their 3′ end which can vary in length from 20 to over 200 bases depending on the species [1]. The 3′-polyadenosine (poly A) tail is necessary for the translation and stability of the mRNA in the cytoplasm through its interaction with poly A binding protein (PABP), the 5′ cap, and the eukaryotic initiation factors (eIF) [2, 3]. In vivo, the poly A tails of mRNA are produced through specific exonuclease trimming of the 3′ end followed by poly (A) polymerase. mRNA that is produced synthetically through in vitro transcription (IVT) gains a poly A tail either by encoding the polyadenosine sequence into the template DNA or by having the polyadenosines added post-synthesis using a polyadenylase. It has been postulated that longer tails increase translation efficiency, provide mRNA stability, and can possibly affect immune response although there has been some recent research suggesting that tail length of mRNA and translation efficiency are linked only during specific cell development stages [4–10]. In the context of therapeutic mRNA which has been designed to produce a specific therapeutic protein or immune-stimulating antigen, the longevity and immunogenicity of the mRNA is important as it influences the duration and/or intensity of protein expression.

Understanding poly A tail length and how it relates to mRNA stability and protein expression requires methods to accurately characterize the poly A tail length of IVT synthesized mRNA. There are many techniques to do this, such as, RNAse H cleavage, chromatographic methods (poly dT or poly U columns), PCR-based assays like LM-PAT and ePAT, as well as next-generation sequencing methods such as TAIL-seq and PAL-seq [4, 6, 11–16]. Next-generation sequencing approaches have the highest resolution (1 base reported for TAIL-seq) whereas the other approaches that use gels or chromatographic methods usually show smears or broad peaks for a population of different tail lengths. The sequencing based

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techniques, however, can be complicated by the indirect nature of their measurement (RNA must be converted to cDNA by amplify long stretches of adenosines which can be difficult), and they also require ligation of streptavidin handles to the mRNA, a step that usually requires overnight incubation. Unlike the above methods, mass spectrometry (MS) is a label-free direct measurement technique that has the ability to distinguish between single nucleotides by their differences in mass. Mass spectrometry has been used to accurately determine the base composition of PCR products up to 500 bases in length, and this approach has even been commercialized into a diagnostic platform for identifying multiple microbial pathogens simultaneously using 120 mer PCR products [17–20]. The capability of MS to directly measure multiple oligonucleotide sequences simultaneously with single nucleotide resolution therefore makes it well suited to study mRNA poly A tail length. Here, we describe a novel method to determine mRNA poly A tail length using LC-MS and apply this method to characterize IVT-synthesized mRNA of 2100 nt and having tails between 27 and 117 nucleotides long.

Method and materials

DNA template preparation

DNA plasmids were generated by conventional cloning methods. Unique restriction enzyme sites were added to the DNA plasmids to allow for the production of poly A-encoded templates or tailess DNA templates. A *Not*I restriction enzyme site was placed prior to the poly A tail, and linearization with this enzyme generated DNA templates lacking the poly A tail. The *Bsp*QI, *Bbs*I, or *Xho*I restriction site was placed at the end of the poly A tail and linearization of the DNA with this enzyme generated DNA templates with poly A tail encoded. T7, SP3, or T3 RNA polymerase promoter sequences were encoded within the plasmids to generate mRNA using different RNA polymerase forms. Linearized DNA templates were analyzed by gel electrophoresis and Sanger Sequencing prior to mRNA preparation.

mRNA preparation

mRNA was prepared using standard run-off IVT procedures and then capped (Cap 0) using the Vaccinia capping system (Part No. M2080S, New England Biolabs, Ipswich MA). Following capping, the mRNA was precipitated with LiCl and then brought up in water. The length and integrity of the mRNA was confirmed with an Experion electrophoresis system (BioRad, Hercules CA) prior to all experiments to ensure there was no degradation (see Electronic supplementary material (ESM) Fig. S1).

For mRNA without plasmid-encoded tails, enzymatic tailing was carried out using *E. coli* poly (A) polymerase and the protocol for mRNA tailing from the manufacturer (Part No. M0276, New England Biolabs, Ipswich MA).

When T3 and SP6 RNAPs were used, their respective promotors replaced the T7 promotor in the sequence. The mRNA DNA template was linearized by either an *Xho*I or *Bbs*I, or *BspQ*I, or *Not*I endonucleases to ensure 3′ ends of mRNA has either only stretches of adenosines (*Bbs*I or *Bsp*QI) or extra sequence after poly A (*Xho*I) or no poly A (*Not*I).

Oligonucleotides used to validate the method were purchased from Integrated DNA Technologies (Coralville IA) and were brought up in DI water and used directly.

T1 cleavage and poly A tail isolation procedure

The cleavage and isolation procedure is similar to one we have used to determine mRNA capping efficiency [21]. One hundred to one hundred twenty picolomes of oligonucleotide or mRNA (previously heated to 95 °C and then quickly cooled to 25 °C) in water was added to an Eppendorf tube along with 10 μl of 10× RNAse H buffer (New England Biolands, Ipswich MA) and 2 μl of RNAse T1 (Life Technologies #AM2280 1000 units/μl) followed by water to make a 1× buffer (1× buffer—5 mM Tris at pH 7.5, 0.5 mM EDTA, 1 M NaCl). The sample was then vortexed briefly and kept at 37 °C for 3 h. After 3 h, 100 μl of the mixture was added to 75 μl of Oligo $(dT)_{25}$ magnetic beads (Dynabeads mRNA Purification Kit, No. 61006, Invitrogen) and isolated according to the manufacturer's protocol. The final rinse step in the protocol was changed to using 100 μl of 100 mM ammonium acetate instead of 100 μl washing buffer in order to remove any sodium that might interfere with the LC-MS analysis.

Using the magnet, the ammonium acetate wash was removed from the beads. Next, 100 μl of 75% methanol (MeOH) that had been heated to 80 °C was added to the beads to release the bound poly A tails. The methanol bead mixture was heated to 80 °C on a hot plate for 1 min and placed on a magnet for 1 min. The supernatant containing the cleavage product and released poly A tail was collected and dried down to 10 μl with an evaporative centrifuge at RT for ~ 45 min. Finally, the sample was resuspended in 50 μl of 100 μM EDTA/1% MeOH for LC-MS analysis.

LC-MS analysis

Standard conditions reported in the literature were used for the LC-MS analysis of oligonucleotides [22–24]. Analysis of the cleaved 3′ mRNA fragment was conducted with an Acuity UPLC (Waters, Milford MA) equipped with a TUV detector that was connected to a QExactive orbitrap (Thermo Scientific, Grand Island, NY). Mobile phase A consisted of 200 mM hexafluoro isopropanol + 8.15 mM triethylamine, pH 7.9, and mobile phase B was 100% MeOH [23]. A Waters Acuity C18 BEH, 2.1 × 100 mm column heated to 75 °C with a flow rate of 300 μl/min was used for all analyses. The gradient profile for elution started at 5% B for 1 min followed by a linear ramp to 25% over 12 min. At 12 min, a 1-min rinse at 90% B began, followed by a return to 5% B at 13 min. UV analysis at 260 nm was conducted online prior to the MS.

All mass spectra were obtained in the negative ion mode, over a scan range of 800–2500 *m/z* at 35,000 resolution. Source and capillary temperatures were set to 350 °C, and the mass spectrometer was calibrated daily with Agilent ESI-L tuning mix (Agilent Technologies, Santa Clara CA) prior to use. All spectra were analyzed using Promass Software (Novatia, Newtown PA) [25]. The deconvolution output mass range was set between 10,000 to 50,000 amu with *m/z* input of 900–2000 *m/z*, and a deconvolution peak width of 1.25 amu.

Protein expression assay

Luciferase expression of mRNA was assessed by monitoring fluorescence in C2C12 cells 24 h post-transfection. Cells were plated at 5000 cells/well 24 h prior to the transfection, and 25 ng of mRNA was transfected into the cells using *TransIT* transfection reagent (Mirus, Madison, WI).

Results

Sample mRNA is first digested with RNAse T1 which cleaves phosphodiester bonds at the 3′ side of guanine. Cleavage fragments containing poly A stretches are then isolated using oligo dT-coated magnetic beads. Magnetic beads functionalized with strands of polythymidine DNA (usually 25 mer in length) are commonly employed for mRNA isolation and use the poly A tail found in mRNA as a handle for capture via Watson-Crick base pairing [26–28]. After washing the beads to remove unbound cleavage products and reaction buffer, the bound poly A species are eluted off the beads and then injected into the LC-MS for analysis. After analysis, the collected electrospray mass spectra are processed and using the known sequences of the mRNA T1 cleavage products are identified by mass.

The method was validated using two synthetic RNAs (CCUGAAAAAAAAAA and CCUGAAAAAAAAAA AAAAAAAAAA) designed to produce a 10- or 20-mer poly A product following T1 digestion. Optimal T1 digestion time was determined by testing a synthetic standard at 1, 3, and 24 h followed by dT isolation and LC-MS analysis. The 1- and 3-h incubations produced a single fragment corresponding to the fully cleaved poly A product and the presence of a single species indicated that complete digestion of the starting oligo was achieved. The 24 h digestion time however, showed significant cleavage of the poly A strand itself (ESM Fig. S2). As a result, the 3-h incubation time was selected over the 1 h to ensure maximum digestion of the mRNA without producing significant nonspecific cleavage.

An equal mixture of a 10- and 20-mer synthetic poly A sequence was tested and capture efficiency depended on tail length with longer tails being captured more efficiently (Fig. 1A). We next tested a mixture of mRNA with 64 and 117 poly A tails and did not observe the large bias in peak areas like that seen for the smaller poly A tails (Fig. 1B) or a change in extraction efficiency from the mRNA alone (not in a mixture). This behavior has been observed by others, and it appears that oligo dT beads have a capture bias for tails greater than around 40 nt [29–31]. Recovery of mRNA with a 117-mer poly A tail by oligo dT beads was determined to be around 70% which is consistent with others' evaluations of magnetic dT beads (ESM Fig. S3) [28]. In order to isolate smaller poly A sequences with lower Tms, it would be necessary to increase the binding strength of the interaction which could be achieved using modified nucleotides such as locked nucleic acids or peptide nucleic acids which have been used to enhance mRNA extraction [32, 33].

Using this method, we examined a 2100-nt-long mRNA with plasmid encoded tail lengths of 27, 64, 100, and 117 polyadenosines along with enzymatically tailed mRNA. In all cases, enzymatically tailed mRNA produced significantly broader distributions of poly A tails than did plasmid-encoded tails (Fig. 2). In some cases, the tail lengths of the enzymatically adenylated mRNAs ranged from 30 to over 120 adenosines. These results are in agreement with those from Holtkamp et al. who also found that enzymatical tailing resulted in a large distribution of tail lengths [34]. Several attempts of enzymatically tailing various mRNAs failed to produce tail length distributions as narrow as plasmid encoding so all work was done using plasmid encoded poly A tails.

The results for plasmid-encoded tails are shown in Fig. 3 which highlights the chromatographic resolution for three tail lengths. The ion-paired reversed phase chromatographic resolution decreases with increasing tail length and single nucleotide resolution is lost between 27 and 64 poly A tail length. It is important to note that chromatographic separation of each distinct tail length is not needed for identification as the mass spectra from co-eluting poly A species can be distinguished from one another by the deconvolution software. A processed or deconvoluted mass spectrum (Fig. 3B) revealed the mass for the expected 100 mer poly A tail (mass 33,163 in box) along with a series of masses separated by 329 amu (± 2 amu) which is the mass of adenosine. The number of different poly A tail species and their relative abundance can be determined by the number and intensity of peaks separated by the mass of adenosine. The observed masses are ± 2 amu from the theoretical 329 difference, and this variance arises from the resolution of the mass spectrometer, however, as the next closest nucleotide in mass is guanosine (G), a 16-amu difference is easily distinguishable. Table 1 shows the masses of the observed and expected T1

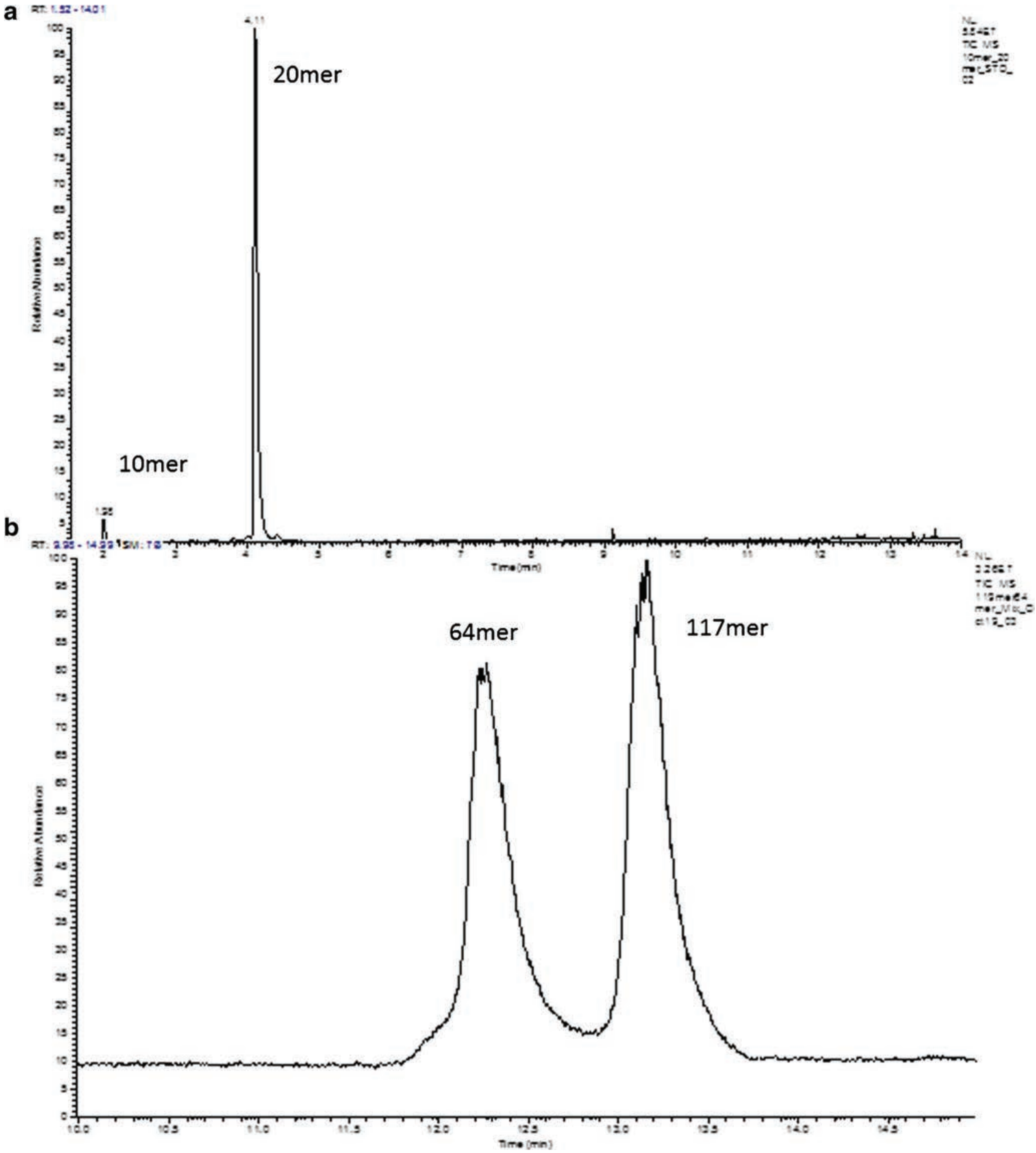


Fig. 1 Example chromatograms of isolated poly A tails. **A** Total ion chromatogram of an equal mixture of the 10 and 20 mer poly A test sequences after 3 h T1 cleavage followed by isolation with oligo dT 25 mer magnetic beads. The total ion chromatogram from the mass spectrometer shows the T1 cleaved product for the 10 and 20 mer poly A sequences. Single peaks for each sequence indicated that digestion was complete after 3 h and that the 10 mer is recovered less efficiently than the 20 mer. **B** Total ion chromatogram of isolated tails from a mixture of two IVT mRNAs, with either a 64 or a 117 length poly A tail

cleavage fragments for the different tail lengths studied. The reproducibility of the method was tested on a 2100-nt mRNA with a poly A tail length of 117. Four separate T1 digestions were put through the procedure and the resulting tail length distributions as well as the mass spectral intensities showed little variation (Fig. 4A).

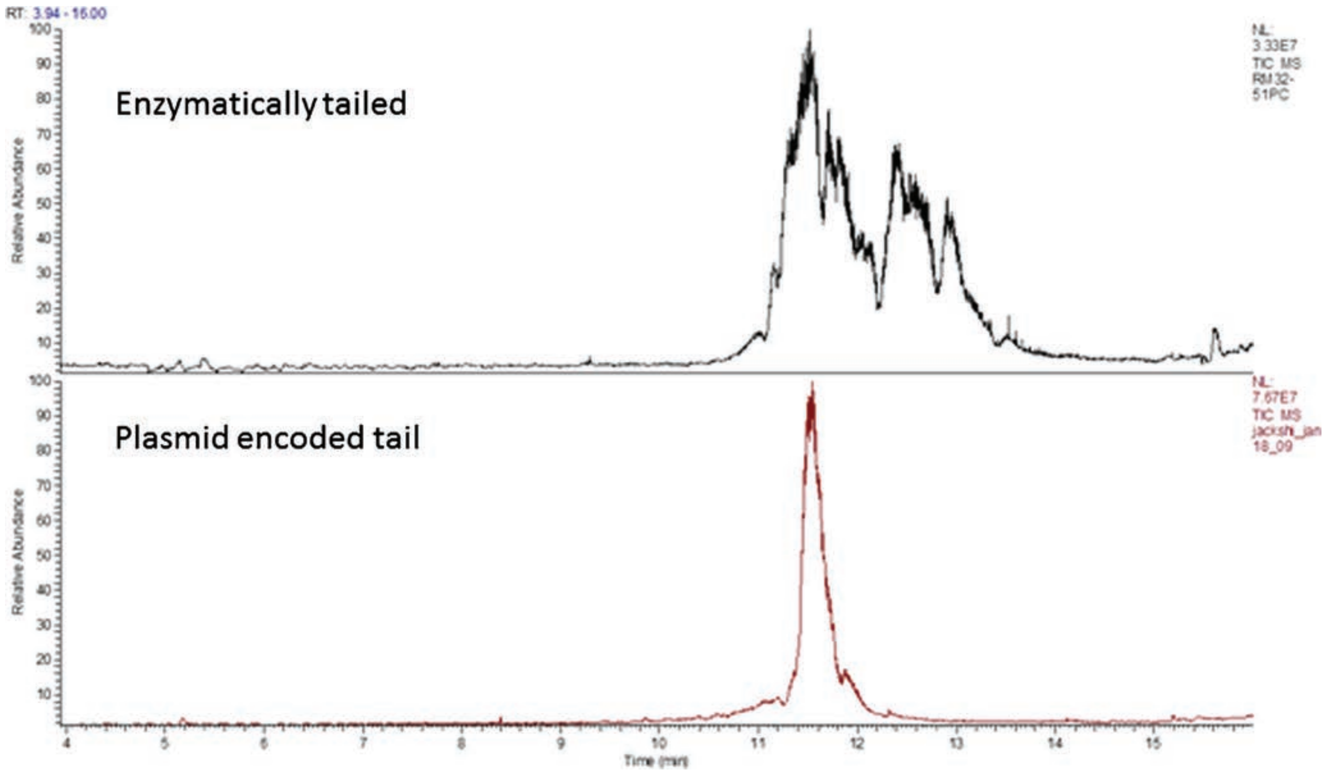


Fig. 2 Total ion chromatograms of poly A tail analysis by LC MS. The top chromatogram of the enzymatically tailed mRNA has a much wider peak indicating a larger dispersity of poly A tail lengths compared with the mRNA with plasmid encoded tails

Poly A tail length distribution

Analysis of mRNAs with different poly A tail lengths revealed that none of the mRNAs had a single unique tail length but contained a distribution of lengths (Figs. 4 and 5). Close inspection of the Sanger sequencing results for the DNA template showed that there are populations of plasmids with different poly A tail lengths. For the plasmid encoding 117 As, the fluorescent traces have a signal for adenosine ranging from positions 109 to 122 (Fig. 4B) despite being called at 117. These minor populations were transcribed into the mRNA, and the observed tail lengths closely matched the adenosine traces in the sequencing. This was also true for plasmids encoding 100 mer tails, however, those encoding 27 mer tails contained tail lengths that were longer than expected from the plasmid (ESM Figs. S4 and S5). In order to examine this in more detail, a new plasmid was constructed encoding a 64 A tail with little background in the Sanger sequence traces to indicate variation in the poly A tail sequence (Fig. 5). Interestingly, the LC-MS analysis still produced lengths greater than what was coded in the plasmid (Fig. 5A). It has been reported that during IVT, T7 polymerase can add non-templated nucleotides to the end of the mRNA so we thought perhaps that the distribution of oligonucleotide sequences differing by one adenosine could be a result of nontemplated addition although unlikely that only adenosine would be selectively added [35]. Analyses of mRNA that was produced using DNA template linearized using *Xho*I restriction site downstream from poly A and therefore did not terminate in A also produced a series of peaks separated by additional adenosines. This observation confirmed that adenosines were not added to the end of the sequence because they would not have remained after the T1 digestion. Together, these observations indicate that the addition (and later observed subtraction) of adenosines is taking place within the poly A tail region and is most likely a result of transcriptional slippage [36–39]. In sequences containing homopolymer tracts of A or U, the RNA strand that is produced as the RNA polymerase transcribes the DNA can slip (either forward or backward) off its DNA template resulting in the addition or subtraction of bases (ESM Fig. S6). Previous work by Wagner et al. demonstrated slippage was most likely to occur in A or T homopolymer tracts at least 10–11 bases in length and that the addition of bases was much more likely than subtraction [36]. Our results are consistent with Wagner's, and the distributions we observed were all biased towards tail lengths longer than templated. In addition, we wondered if tail length distribution might be related to ATP concentration so the 27, 64, 100, and 117 mer tail length mRNAs were made with ATP concentrations 2× and 4× over normal levels (12 and 24 mM). Increasing the ATP concentration, however, did not alter the distribution of any tail length which is a similar result to MacDonald who noted little change in tail length distributions with T7 RNAP at lower NTP concentrations [38].

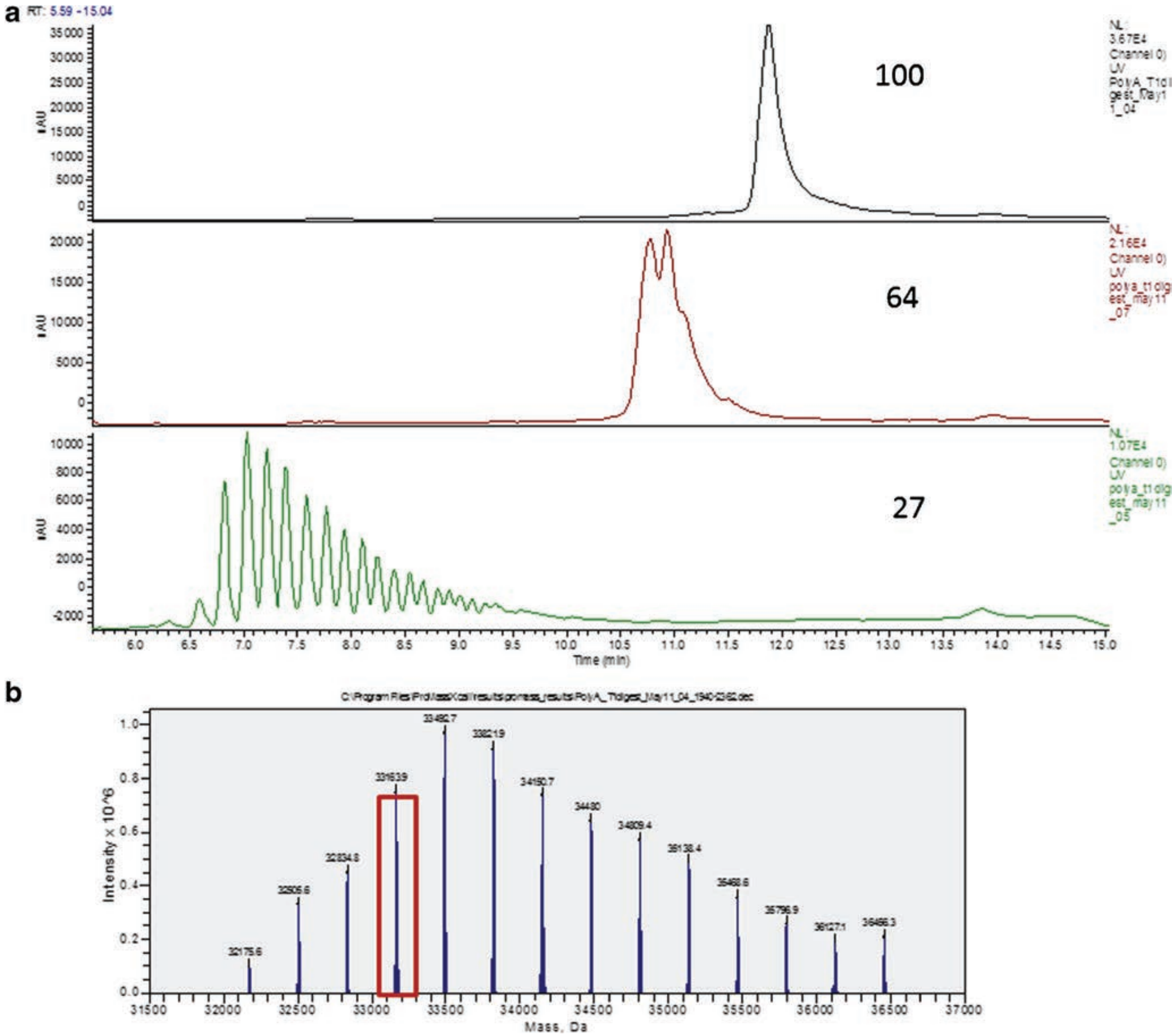


Fig. 3 **A** UV 250 nm chromatograms from the LC MS analysis of mRNA with poly A tails of 27, 64, and 100. At the 27 mer length, each individual peak represents a single tail species differing by one adenosine. This single nucleotide resolution is lost at the 64 mer length where peaks coalesce. **B** The resulting deconvoluted electrospray mass spectrum of the single peak in the chromatogram for the 100 mer tail length. Mass versus spectral intensity is plotted, and the series of peaks separated by the mass of adenosine represents the different poly A tail lengths observed. The peak at mass 33,163.9 outlined in the box represents the mass of the 100 mer poly A tail that is the plasmid encoded length

Table 1 Theoretical and experimentally observed masses of various poly A tails after T1 digestion

Tail length (CA_n)	Calculated molecular weight (avg)	Observed molecular weight
117	39,759.7	38,760.6
100	33,164.1	33,163.9
64	21,312.6	21,312.2
27	9131.8	9130.8

RNA polymerases and poly A tail length distribution

Work by Molodtsov et al. demonstrated that the mRNA sequence as well as the RNAP can affect the amount of slippage that occurs [37]. To test this idea ourselves, mRNA poly A tail lengths of 64 adenosines were in vitro-transcribed using either T7, T3, or SP6 RNAPs. These sequences were all found to have distinct poly A tail regions with no significant background A signal in the Sanger sequencing traces (ESM Fig. S7). In addition, the mRNA was made using a template DNA linearized using either a *Bbs*I or an *Xho*I restriction

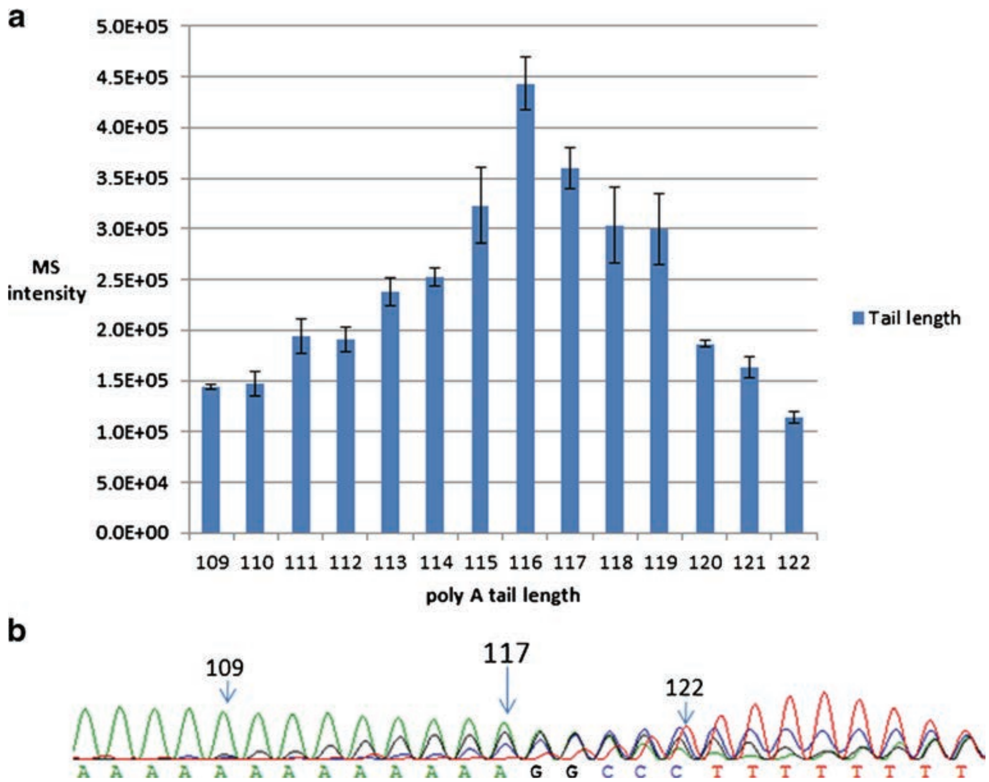


Fig. 4 **A** Graph of average mass spectral signal intensity versus poly A tail length for four separate T1 digestions and analyses of a mRNA with a tail length of 117 As. **B** Sanger sequencing results of the 3′ end of the plasmid used to synthesize this mRNA. The fluorescent intensity traces for each base along with the called sequence are shown. Adenosine is designated by the green trace and the numbers above the sequence indicate the number of As in the tail. This sequence was called at having 117 As, but close inspection shows that other bases are present which decrease the number of As in the tail to 109 and that additional As are present out to 122

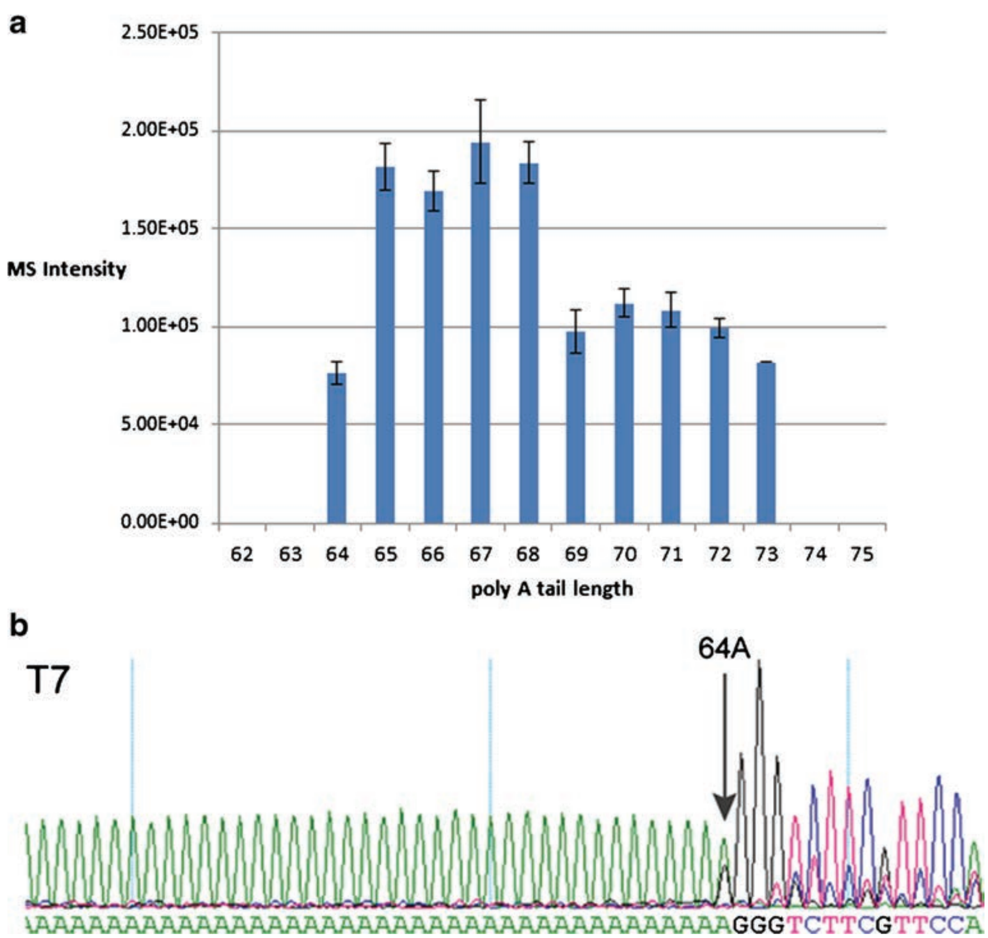


Fig. 5 **A** Graph of average mass spectral signal intensity versus poly A tail length for a mRNA with a tail length of 64 As ($n = 4$). **B** Sanger sequencing results of the 3′ end of the plasmid used to synthesize this mRNA. The fluorescent intensity traces for each base along with the called sequence are shown. Adenosine is designated by the green trace and inspection of the trace shows that there are no additional As after the designated 64

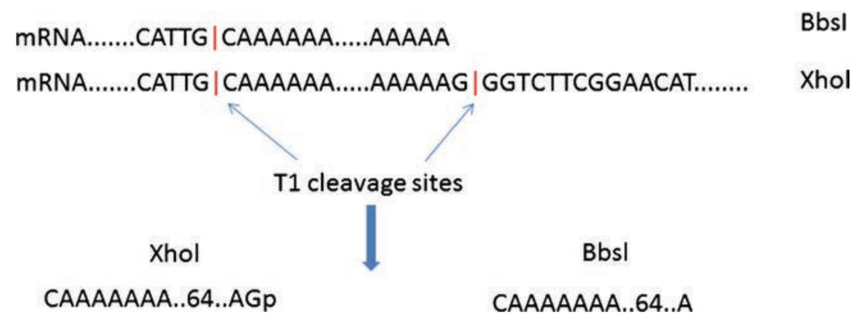
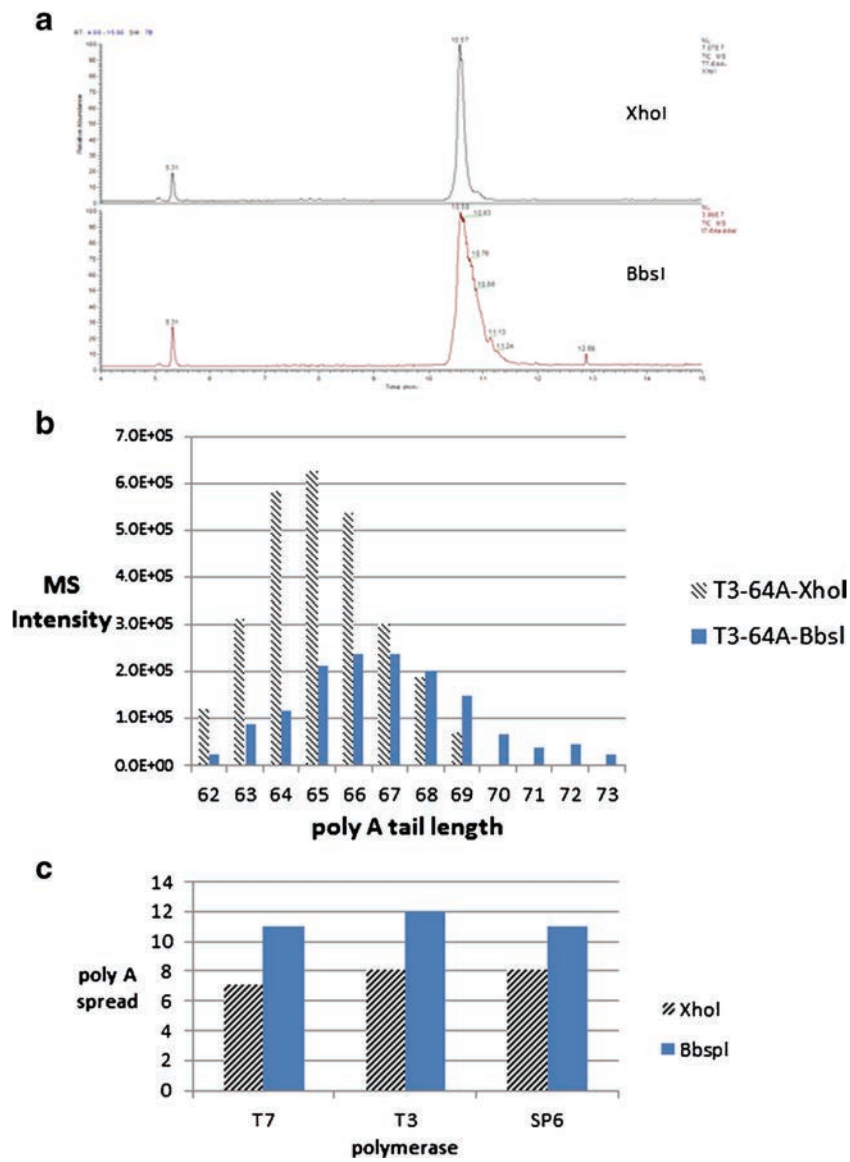


Fig. 6 mRNA was synthesized from DNA containing either a *Bbs*I or *Xho*I restriction site. The *Bbs*I produces mRNA with a 3′ poly A tail ending in adenosine whereas mRNA made with the *Xho*I site have a short sequence after the poly A. T1 cleavage of the tails from the mRNA produces sequences that either end with a 3′ adenosine or with a guanosine and 3′ phosphate

endonuclease, which produced mRNA with either a 3′ poly A tail ending in adenosine (*Bbs*I) or ending in a stretch of additional sequence after the 64 poly A tail (*Xho*I) (Fig. 6). According to the hypothesis put forward by Molodtsov et al. the addition of sequence at the end of the homopolymeric region will limit the amount of slippage [37]. Our observations were consistent with this hypothesis and *Xho*I mRNAs did show a narrower distribution of poly A tails than *Bbs*I

Fig. 7 **A** Total ion chromatograms comparing the 64 mer poly A tails from *Bbs*I and *Xho*I mRNA sequences showing a narrower distribution in tails from *Xho*I. **B** Plot of tail length versus mass spectral intensity for *Bbs*I and *Xho*I mRNA. The distribution of tail lengths from *Bbs*I is greater than *Xho*I and *Bbs*I has more tails of greater than the templated length. *Xho*I tail length distribution is narrower and centered around the templated number. **C** Distribution of poly A tail length for *Bbs*I and *Xho*I mRNA made with either T7, T3, or SP6 RNAP



regardless of which RNAP was used (Fig. 7A, B). In fact, the type of RNAP used did not make a significant difference in any of the observed poly A tail distributions (Fig. 8C). The narrower distribution caused by the introduction of a sequence on the 3′ end is noteworthy because it indicates that most of the slippage is taking place near the end of the poly A tail. Interestingly, the intensity of the *Xho*I tails is also greater which indicates more mRNA is being made at these lengths than *Bbs*I whose intensity is spread out over a greater distribution (Fig. 7B). Unfortunately, leaving the *Xho*I sequence on the 3′ end of mRNA has been shown to significantly reduce mRNA protein expression [40]. In work by Grier et al., it appears that having one or two guanosines on the end of the poly A tail, however, does not negatively impact protein expression but we do not know how this impacts tail length distribution [41].

Another variation of these sequences was also made, where a single C was introduced at position 7 from the start of the poly A tail. This should disrupt any slippage from occurring as it limits the tract of poly A tails to less than the number reported to cause slippage. Introduction of the single C at the start of the tail however, did not change the tail length distributions for any of the RNAPs or restriction sites (results not shown). In hindsight, this result is not surprising as the stretch of poly As (now at 57) is long enough to provide plenty of opportunity for slippage to occur.

Translation efficiency

mRNA poly A tail lengths have been studied to determine an optimal length for maximum protein expression [34, 40–42]. In our examination of the literature, there does not appear to be a universal tail length that yields a maximum protein expression for all cell types and mRNAs. For example, Holtkamp et al. found that 120 As gave the highest yield, Elango 60 As,

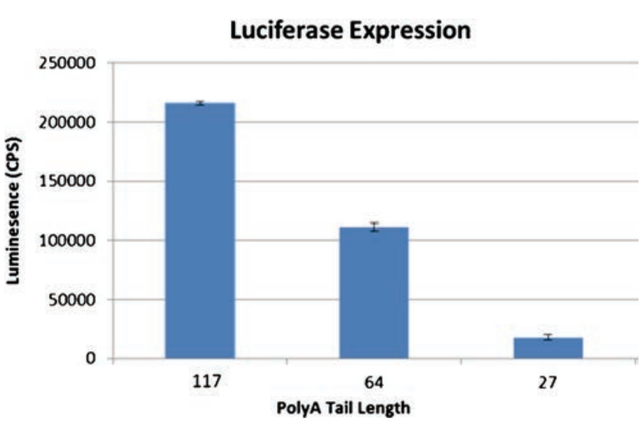


Fig. 8 Luciferase expression of mRNA. C2C12 cells were plated at 5000 cells/well 24 h prior to the transfection. Twenty five nanograms of WT mRNA were transfected into the cells, and luciferase expression was checked at 24 h post transfection

and Mockey observed that 100 As were optimal. It is hard to draw a conclusion because each of these studies used different mRNA sequences and cell types for expression. In addition, some of these studies used enzymatically tailed mRNA which could contain large distributions in tail lengths [34]. For our particular set of circumstances, however, we observed that protein expression increased with increasing tail length and was highest with the longest tail produced (117 A) (Fig. 8).

Conclusion/discussion

The 3′ poly A tail has been shown to affect mRNA function in a variety of ways and measurement of the tail length in IVT-synthesized mRNA can be helpful in understanding how length is related to function. Assessing the heterogeneity of poly A tails in IVT mRNA can also be important from a clinical standpoint as mRNA is increasingly being produced for therapeutic purposes and having a consistent and well-defined product is important to reproducible activity. In order to characterize IVT mRNA, we developed a simple and rapid LC-MS-based method for directly (no conversion to cDNA) determining poly A tail length with single-nucleotide resolution.

Tail lengths can be analyzed in less than 5 h with most of that time spent incubating for T1 cleavage. We have used magnetic oligo dT beads which are a common technique for isolating mRNA, and they worked well to capture cleaved poly A tails but did show a bias for capturing longer tails in a mixed population. This bias did not appear to be present for tails around 64 nt and above. Analysis of the isolated poly A tails employed standard LC-MS conditions for oligonucleotides and used routine MS data processing to calculate the masses of the tail species. The upper limit of the length of tail that can be analyzed, the resolution, and the sensitivity of that measurement are dependent on the mass spectrometer. As mentioned earlier, sequences up to 500 nucleotides in length have been analyzed successfully so this is well within the reported range of mRNA poly A tail lengths [17]. Using the theoretical mass to identify tail length does require knowledge of the 3′ sequence in order to calculate the mass of the T1 cleavage product which would prevent it from being used to determine the tail length of unknown mRNAs. Although even without sequence information, the distribution of tail lengths could be obtained from the number of peaks separated by the mass of a nucleotide.

Our first observations demonstrated that poly A tails encoded in the DNA plasmid produced much narrower poly A tail distributions than mRNA that was enzymatically tailed post-IVT. This was a finding previously noted by Holtkamp et al. and as a result, we chose plasmid-encoded tail lengths to demonstrate our application [34]. Using plasmid-encoded poly A tails of 27, 64, 100, and 117 in length, we found that the distribution of tail lengths closely matched the sequence populations observed in Sanger sequencing. However, when no variation in tail length was detected in the encoding plasmid, tails greater than the DNA template were observed. The variation in tail length unrelated to the plasmid sequence was attributed to transcriptional slippage by the RNAP. Different RNAPs did not alter tail length distributions and we found that T7, T3, and SP6 all produced the same amount of slippage. It is possible that mutated or other types of RNAPs could narrow the poly A tail distribution, for example it has been shown that human mitochondrial RNAP has less of tendency to slip than SP6 [37].

The findings here indicate that tying a specific tail length to the amount of protein expression or other mRNA attribute is problematic particularly if enzymatic tailing is used. This could explain some of the variation between studies looking at optimizing poly A tail length and protein expression. For our specific system, however, we found that protein expression increased up to the maximum 117 poly A tail length that was studied.

The bias in dT extraction efficiency will prevent the use of the method for determining the absolute abundance of each tail species in a mixture that contains tails below 40–60 nt, but these species would still be observed qualitatively to indicate the nature of the different tail species present in the sample. In this regard, it will suffer the same limitations as other tail length profiling methods that use dT extraction to assess a mixed population of small and large poly A tail lengths. An obvious adaptation, and one we are working on, is the combination of PCR-Ligation techniques that do not require oligo dT for isolation followed by analysis with MS. However, in the framework of IVT-produced mRNA with well-defined tail lengths, this method offers a quick way of accurately checking tail length which should be useful to those studying IVT mRNA.

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Compliance with ethical standards

Conflict of interest The authors declare they have no conflict of interest.

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